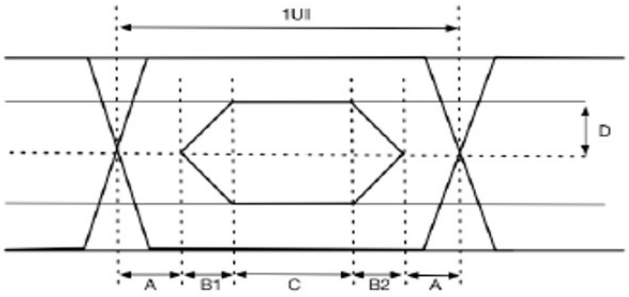
IP DESIGN CHECKLIST

Ramschip,Inc Confidential

1. BASIC INFORMATION	
Process	**SF**
Project name	**Voyager (rams**
Customer	**SE**
Application	**Dis**
Project manager	**E**
IP design classificaiton	**New desig**
QA manager	**Ma**
Target Design review	**YY**
MTO Schedule	**Nov.1.2024 (FEOL Dry GDS)**

2. DESIGN CHECK SHEET

CATEGORY	ITEM
Customer request	
Customer request	
Customer request	
Customer request	
Customer request	**Note:** 1) Termination is 100ohm. 2) VTX_EYE is at least 70% of set current times 100ohm. 3) The jitter measurement is done with second-order clock recovery function and a closed loop tracking bandwidth of 5MHz with a dammping 1.5 4) Jitter specification is guaranteed with 25mVpp ripple on 0.7125V mnimal VDDTX supplies (1MHz ~ 5MHz range) 5) Vod is differential voltage swing, Vod = IB-tx * (R-tx_term * R* rx-term) / (R-tx_term + R-rx_term), IB-tx is programmable Tx driving current specified in next section.
Customer request	
Customer request	
Customer request	
Customer request	
Customer request	

Customer request	**TX Electrical Charateristics**
Customer request	**TX Electrical Charateristics**
Customer request	**TX Electrical Charateristics**
Customer request	**TX Electrical Charateristics**
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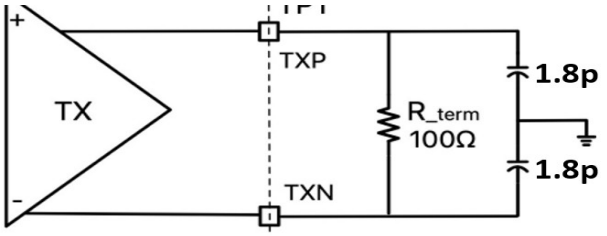
Customer request	**TX Electrical Charateristics**
Customer request	**TX Electrical Charateristics**
Customer request	**TX Electrical Charateristics**
Customer request	**TX Electrical Charateristics**
Customer request	**TX Jitter** Note: TX jitter is measured with second-order clock recovery function with a closed loop tracking bandwidth defined below and a damping factor of 1.5: BW = 5MHz with SSC off. Note: The jitter specification is guaranteed with supply peak-to-peak noise within 25mV(@2MHz) with 0.7125V at TCON VDD_TX package pin for noise frequency lower than 10MHz, and within 30mVpp(@10MHz) for noise frequency higher than 10MHz
Customer request	
Customer request	
Customer request	**TX Termination** **Figure 2 TX Measurement Termination** TP1
Customer request	
Customer request	

Customer request	
Customer request	
Customer request	
Customer request	
Customer request	
Customer request	TX Programmable Driver Current (Coarse) REG_IB_CORSE[1:0] (Shared with all ports)
Customer request	
Customer request	
Customer request	
Customer request	TX Programmable Driver Current (Fine) REG_IB_FINE[3:0] (bit [3:2] Shared with all ports And bit[1:0] per port)
Customer request	
Customer request	
Customer request	Agile Clock System
Customer request	
Customer request	
Customer request	
Customer request	
Customer request	
Customer request	
Customer request	
Customer request	Power sequence [Power Up / Down]
Customer request	Low Power Mode in VBP refer to the "2. Low power mode in VBP" tap
Customer request	
Customer request	
Customer request	
Customer request	PLL input reference clock
Customer request	
Customer request	

Customer request	SSC (output data Spread Spectrum Clocking for low EMI)
Customer request	
Customer request	
Customer request	Symbol clock speed
Customer request	Lane Configuration
Customer request	
Customer request	
Customer request	Termination
Customer request	BIST
Customer request	
Customer request	DFT interface
Customer request	SCAN CLOCK
Customer request	APB Clock
Customer request	P/N Swap
Customer request	Jitter Injector
Customer request	Slew Control
Internal	LDO
Internal	
Internal	
Internal	
Internal	STA

Internal	**Driver**
Internal	
Internal	
Internal	
Internal	
Internal	
Internal	
Internal	
Internal	
Internal	

	C
4nm LPP	
schip internal title)	Customer spec &
:C S.LSI	Internal check
splay TX	
3J. Lee	
n / Modification	
ark. Kim	
YMMDD	
) . Nov.15.2024(FEOL Final GDS)	

PARAMETER[Symbol] / CHECK POINT	
UI interval check [UI]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 2
[Slope On / SSC Off] Differential Mode output jitter [A*2]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope On / SSC Off] Eye shape [B1]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope On / SSC Off] Eye shape [B2]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope On / SSC Off] Eye shape [C]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope On / SSC Off] Eye height [D]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope Off / SSC Off] Differential Mode output jitter [A*2]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope Off / SSC Off] Eye shape [B1]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope Off / SSC Off] Eye shape [B2]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope Off / SSC Off] Eye shape [C]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0
[Slope Off / SSC Off] Eye height [D]	● 0.85 ~ 2.1Gbps : ● 2.1 ~ 4.0Gbps : 0

Differential Peak-to-peak Output Voltage Swing [VDIFFp-p] Comment: <±20%, step size <50mV, With 100ohm termination and 100% bias setting	Min: 0.1V / Typ: 0.3V /
Minimum TX Eye Width at Tx package pins [TTX-EYE_CHIP]	Min: 0.65 UI
Maximum time between the jitter median and maximum deviation from the median at Tx package pins [TTX-EYE-MEDIAN-to-MAX-JITTER CHIP] Comment: After CDR Filter (5MHz, Damp 1.5, Second Order)	Max: 0.175UI
TX output common mode voltage control VCM definition: | Vop+Von|/2" [VCM] Comment: (<± 10%)* VCM level can changed up to supply voltage condition. Step size: 12.5mV	Min: 0.3V / Typ: 0.375'
Variation in VCM between 0 and 1 [ΔVCM]	Max: 14 mVrms
Variation in VOD between 0 and 1 [ΔVOD]	Max: 30mV
TX Short Circuit Current Limit [ITX-SHORT] Comment: Total drive current of the transmitter when it is shorted to its ground.	Max: 50mA
Differential Return Loss at 0.45Gbps at TX Package pins [RLTX_DIFF] Comment : Piecewise linear loss line between the specified points.	Max: -12dB
Differential Return Loss at 0.9Gbps at TX Package pins [RLTX_DIFF] Comment : Piecewise linear loss line between the specified points.	Max: -9dB
Differential Return Loss at 1.8Gbps at TX Package pins [RLTX_DIFF] Comment : Piecewise linear loss line between the specified points.	Max: -6dB
Differential Return Loss at 3.0Gbps at TX Package pins [RLTX_DIFF] Comment : Piecewise linear loss line between the specified points.	Max: -4dB
TX Output Capacitance [CTX_OUTPUT]	Max: 2pF

Termination resistance [ZTX-DIFF-DC]	Min: 75 Ohm / Typ: 90
D+/D- TX Output Rise/Fall Time at Tx package pins when Slope On is enabled [TTX-RISE / TTX-FALL] Comment: At 20%-to-80% with external termination 100ohm and external cap 1.8pF	Min: 50 ps / Max 170 p
D+/D- TX Output Rise/Fall Time at Tx package pins when Slope On is disabled [TTX-RISE / TTX-FALL] Comment: At 20%-to-80% with external termination 100ohm and external cap 1.8pF	Min: 50 ps / Max 140 p
Intra-lane Output Skew at TX package pins [TTX-SKEW-INTRA-LANE] Comment: Informative to meet ΔVCM.	Max: 10 ps
Inter-lane (within port) Output Skew at TX package pins [TTX-SKEW-INTER LANE] Comment: Max 50ps with programmability	Max: 50 ps
Lane Intra-pair Rise-fall Time Mismatch at TX package pins. [TTX-RISE_FALL-MISMATCH] Comment: D+ rise to D- fall mismatch and D+ fall to D- rise mismatch.	Max: 10%
[Slope on] **VActive, 25mVpp ripple** **VBlank, 25mVpp ripple** **Transition between VActive and VBlank, 25mVpp ripple** **[TX_Dj]**	0.28UI
[Slope off] **VActive, 25mVpp ripple** **VBlank, 25mVpp ripple** **Transition between VActive and VBlank, 25mVpp ripple** **[TX_Dj]**	0.24UI
[Slope On/Off] **VActive, 25mVpp ripple** **VBlank, 25mVpp ripple** **Transition between VActive and VBlank, 25mVpp ripple** **[TX_Rj]**	0.11UI
[Setting] **111**	75 Ohm
[Setting] **110**	82 Ohm
[Setting] **101**	90 Ohm

[Setting] 100	100 Ohm
[Setting] 011	111 Ohm
[Setting] 010	128 Ohm
[Setting] 001	150 Ohm
[Setting] 000	180 Ohm
[Setting] 00	1 mA
[Setting] 01	2 mA
[Setting] 10	3 mA
[Setting] 11	4 mA
[Setting] 0000	56%
[Setting] 0001 ~ 1110	Linearly interpolate between 56% to 150%
[Setting] 1111	150%
Agile Clocking Change	Up to ± 1.5%
Spread Spectrum Clocking	Up to ± 2%
Modulation scheme	Center spreading
Modulation profile	Triangular
Modulation rate	±0.25% , ±0.5%, ± 0.
Modulation Frequency	10KHz , 20KHz, 30KHz
PLL Lock Time	Max 100us (When PLL Max 30us (When PLL i
Refer to the "1. Power sequence" tap	OK / NG
Normal mode + lane on/off	wake-up time 3us for e
Low power mode Type1	wake-up time 3us
Low power mode Type2	wake-up time 50us
Low power mode Type3	fast wake-up mode 20
Reference clock frequency as PLL input clock	27MHz from X-tal
Duty spec	45% ~ 55%
Total jitter spec	TJ ± 200ps

Modulation Frequency	Max 40KHz
Modulation ratio	Max ±2%
Modulation deviation	Center spread
Symbol clock frequency	4Gbps 기준: 4GHz/12
Number of Lanes	1 common and Max 16
Activated-lane combination	Lane selectable, 16 lan (2/4/6/8/10/12/14/16 la
Lane Skew	Inter lane skew -max 1 UI @ <=12 lan -max 1.3 UI @ >12 lar
Termination resistance	75/82/90/100/111/128
Built in self test	Support
BIST pattern	PRBS7, PRBS12, User Pattern
Design for test interface	SPI
Shift / Capture clock	100MHz
APB operation clock	300MHz
P/N swap function support	Support
Periodic jitter injection function	Support
Slope on/off feature to reduce the EMI effect	Support
Drop out voltage 입력전압 1.8V 의 min 1.53V max 2.07V 에서 LDO의 출력 1.2V가 출력되는지 Check @-40C / 125C	입력전압의 +/- 15% rang
Load regulation 정격 부하 전류의 +-20%에서 LDO의 출력 전압 1.2V가 출력되는지 Check	설계 spec
Power supply rejection ratio 입력전압 1.8에 Ripple +-20% 를 injection 하여 출력 전압 1.2V 가 정격 출력되는지 Check	설계 spec
Current limiting & short circuit protection LDO가 과전류 발생 시 보호기능 수행 여부. 과부하나 단락 발생시 회로를 보호하는 기능 구현 여부	설계 spec
Timing check Setup / Hold time이 요구사항에 충족되었는가?	설계 spec

Output impedance matching - 드라이버 출력 임피던스가 송,수신선의 특성 임피던스에 잘 맞추어져 있는가? -. 임피던스 불일치로 인한 신호 반사 및 왜곡이 발생하지 않은가?	설계 spec
Signal integrity -. 고속 신호 전송에서 반사,왜곡,Jitter가 적정 수준인가? -. Crosstalk 와 EMI가 허용범위 내에 있는가? -. 신호 경로상에 Jitter 및 Skew가 적정 범위내에 있는가?	설계 spec
Driver swing -. 출력 전압의 Swing이 설계 목표 및 option에 따라 맞게 출력 및 조정되는가?	설계 spec
Power consumption -. 드라이버의 전력 소모가 Targeting power를 만족하는가? -. Sleep mode 및 동적 전력 제어가 구현되었는가?	설계 spec
Drive strength -. Driver가 외부 부하(cable/수신회로)를 충분히 구동 시킬 수 있는가? -. 과부하 또는 단락 시 보호 기능이 활성화 되어 있는가? -. Drive strength option은 Target대로 작동하는가?	설계 spec
Differential signal integrity -. 차동 신호의 대칭성 및 신호 무결성이 유지되는가? -. Noise 발생은 허용치 이내인가?	설계 spec
EMI / EMC -. Driver에서 발생하는 EMI는 허용된 수준 이내인가? -. 적절한 EMI Filtering 또는 Shielding이 적용되었는가? -. 인접회로에 대한 EMC가 확보되었는가?	설계 spec
Driver bandwidth -. Driver가 전송할 신호 대역폭을 충분히 제공하는가? -. 고속 신호 전송 시 주파수 응답이 적절한가?	설계 spec
Driver linearity -. Driver의 출력이 선형적으로 변화하여 왜곡 없이 신호가 잘 전달 될 수 있는가? -. 출력에서 Distortion 없이 신호 품질이 유지되는가?	설계 spec
Output current & voltage clamping -. Driver 출력 전류 및 전압 제한 (Claming)이 설정되었는가? -. 단락 및 과전류 시 보호 동작이 구현되었는가?	설계 spec
Parastic effect -. 기생 인덕턴스 및 케페시턴스가 최소화 되었는가? -. 고속 신호에서 기생효과 신호에 미치는 영향은 적절한 수준인가?	설계 spec

OLOR DESCRIPTION (ITEM)	
Item	PASS
request	NG
	N.A
	Not yet
	SCORE

SPEC Criteria	CHECK RESULT	Check engineer
476 ~ 1176 ps 50 ~ 476 ps	PASS	BJ
0.35UI .39UI	PASS	BJ
0.28UI .40UI	PASS	BJ
0.28UI .21UI	PASS	BJ
0.09UI UI	PASS	BJ
0.7 * VOD mV .7 * VOD mV	PASS	BJ
0.35UI .39UI	PASS	BJ
0.28UI .40UI	PASS	BJ
0.28UI .21UI	PASS	BJ
0.09UI UI	PASS	BJ
0.7 * VOD mV .7 * VOD mV	PASS	BJ

Max 0.4V	**PASS**	BJ
	PASS	BJ
	PASS	BJ
V / Max 0.4V	**PASS**	BJ
	PASS	BJ
	PASS	BJ
	PASS	BJ
	PASS	BJ
	PASS	BJ
	PASS	BJ
	PASS	BJ
	PASS	BJ

Ohm / Max 180 Ohm	**PASS**	BJ
ɔs	**PASS**	X
ɔs	**PASS**	BJ
	PASS	BJ
	PASS	BJ
	NG	BJ
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	

	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	BJ
	PASS	BJ
	PASS	BJ
	PASS	BJ
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
75%, ±1%, ± 2%	**PASS**	
, 40KHz with +/- 5% error	**PASS**	
is shut down during Vblank)	**PASS**	
is not shut down during Vblank)	**PASS**	
	PASS	
each lane	**PASS**	
	PASS	
	PASS	
0us, deep sleep mode 500us	**PASS**	
	PASS	
	PASS	
	PASS	

	PASS	
	PASS	
	PASS	
	PASS	
5 lanes	PASS	
nes target anes configurable)	PASS	
es nes(16 lanes)	PASS	
/150/180 ohm	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	PASS	
	Defer	
ge에서의 LDO 출력 Check	Not yet	
	Not yet	
	Not yet	
	Not yet	
	PASS	

	PASS	
	PASS	
	Not yet	
	Not yet	
	Not yet	
	Not yet	
	Not yet	
	Not yet	
	Not yet	
	Not yet	
	Not yet	

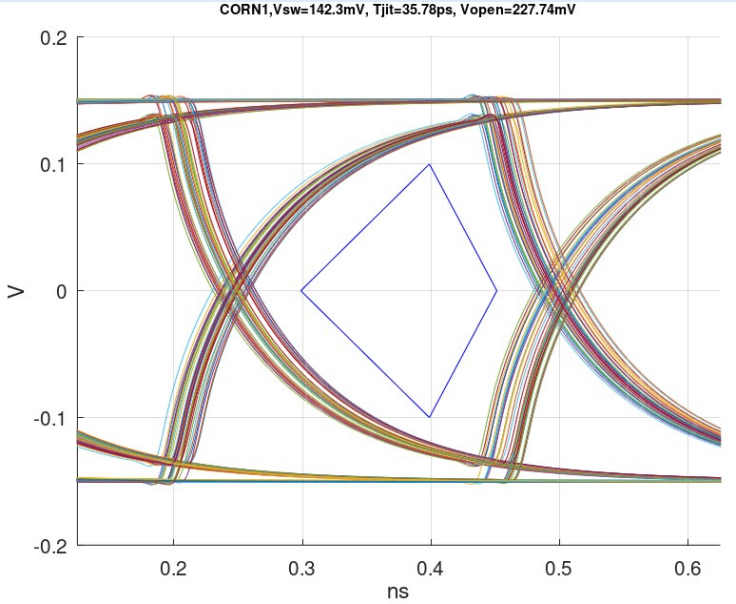
ram-sCHIP
FNALOGIXEDIGNAL

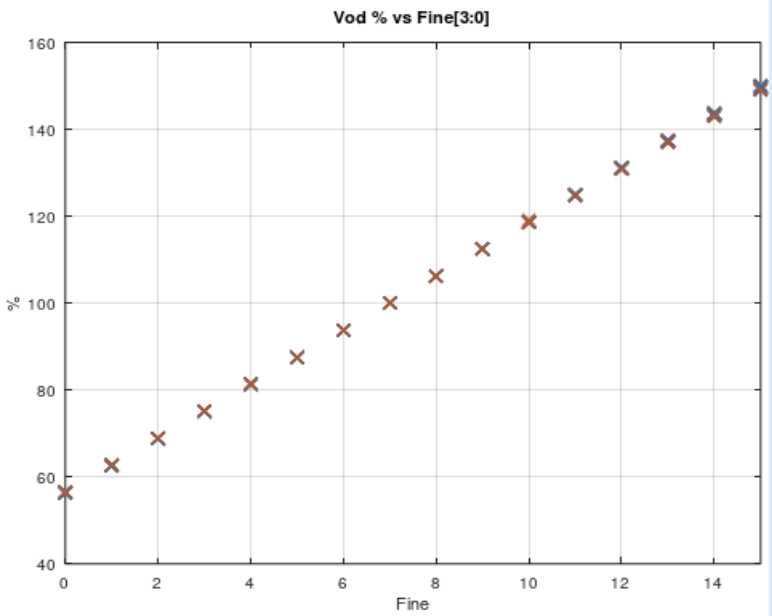
SCORE
80
1
0
13
85%

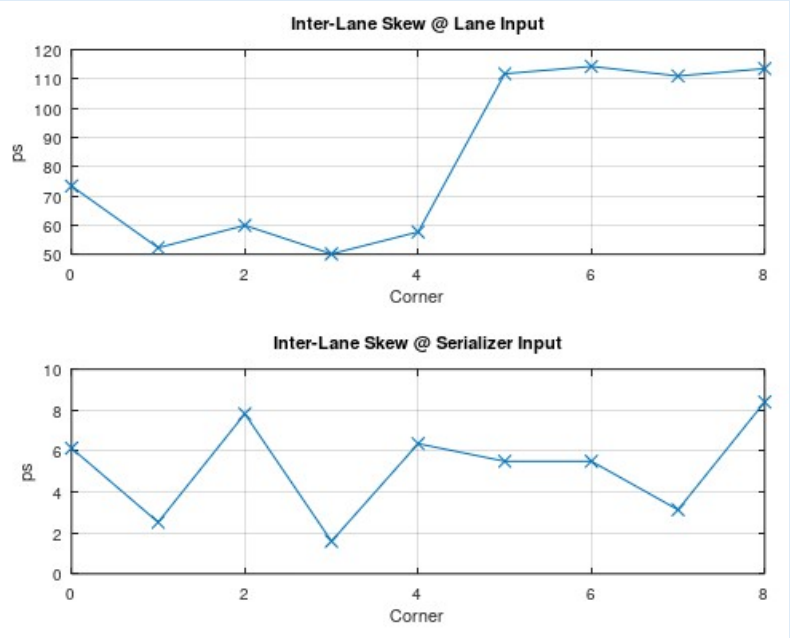
VER. 1.0

PRODUCT (PROJECT)	
NAME/DATE (YY.MM.DD)	NOTE
11/25/2024	DR2 자료 참고
11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	

11/25/2024	
11/25/2024	
11/25/2024	
11/25/2024	σ<4.12mV when VCM=375mV
11/25/2024	rms < 0.2mV when VCM=0.375, IDRV=3mA
11/25/2024	< 1mV
11/25/2024	17.4mA
11/25/2024	-15.6
11/25/2024	-12.6
11/25/2024	-7.4
11/25/2024	-4.3
11/25/2024	1.4pF

11/25/2024	
11/25/2024	
11/25/2024	
	>131ps skew control range w/ 6-16ps step (Total 21 steps)
11/22/2024	>131ps skew control range w/ 6-16ps step (Total 21 steps)
11/22/2024	10.8% (Waive)
	Worst Case Eye 
	Tolerance (+/-15%)

	Max/Min=1.044/0.989
11/23/2024	Max/Min=2.046/1.982
11/23/2024	Max/Min=3.060/2.975
11/23/2024	Max/Min=4.080/3.958
11/23/2024	
	PLL DR 자료
	Logic DR
	PLL DR 자료

	PLL DR 자료
	
	Logic DR
	추후 설계 보완 예정
	설계 Spec 대비 Verificaiton결과 기록
	설계 Spec 대비 Verificaiton결과 기록
	설계 Spec 대비 Verificaiton결과 기록
	설계 Spec 대비 Verificaiton결과 기록
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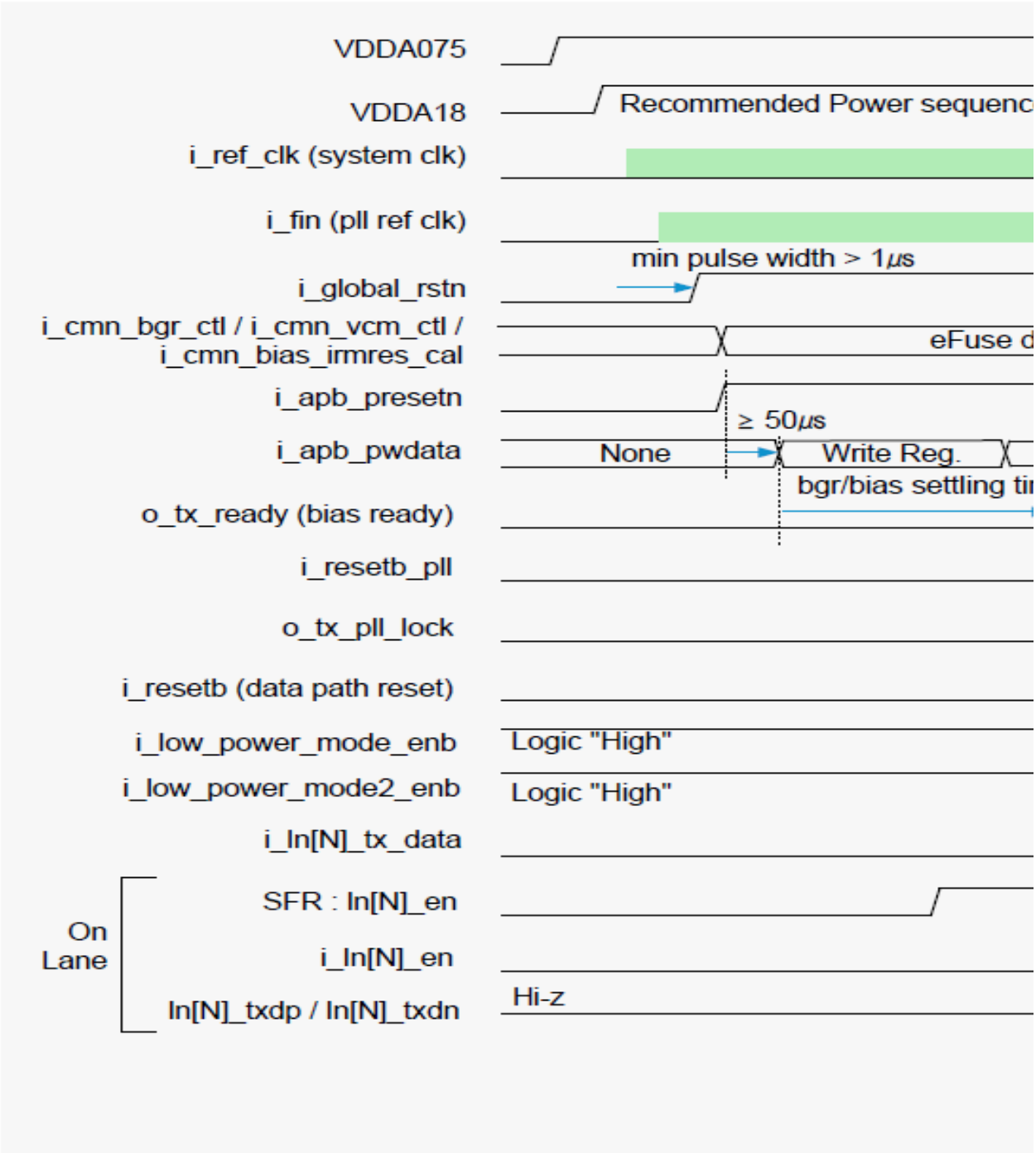
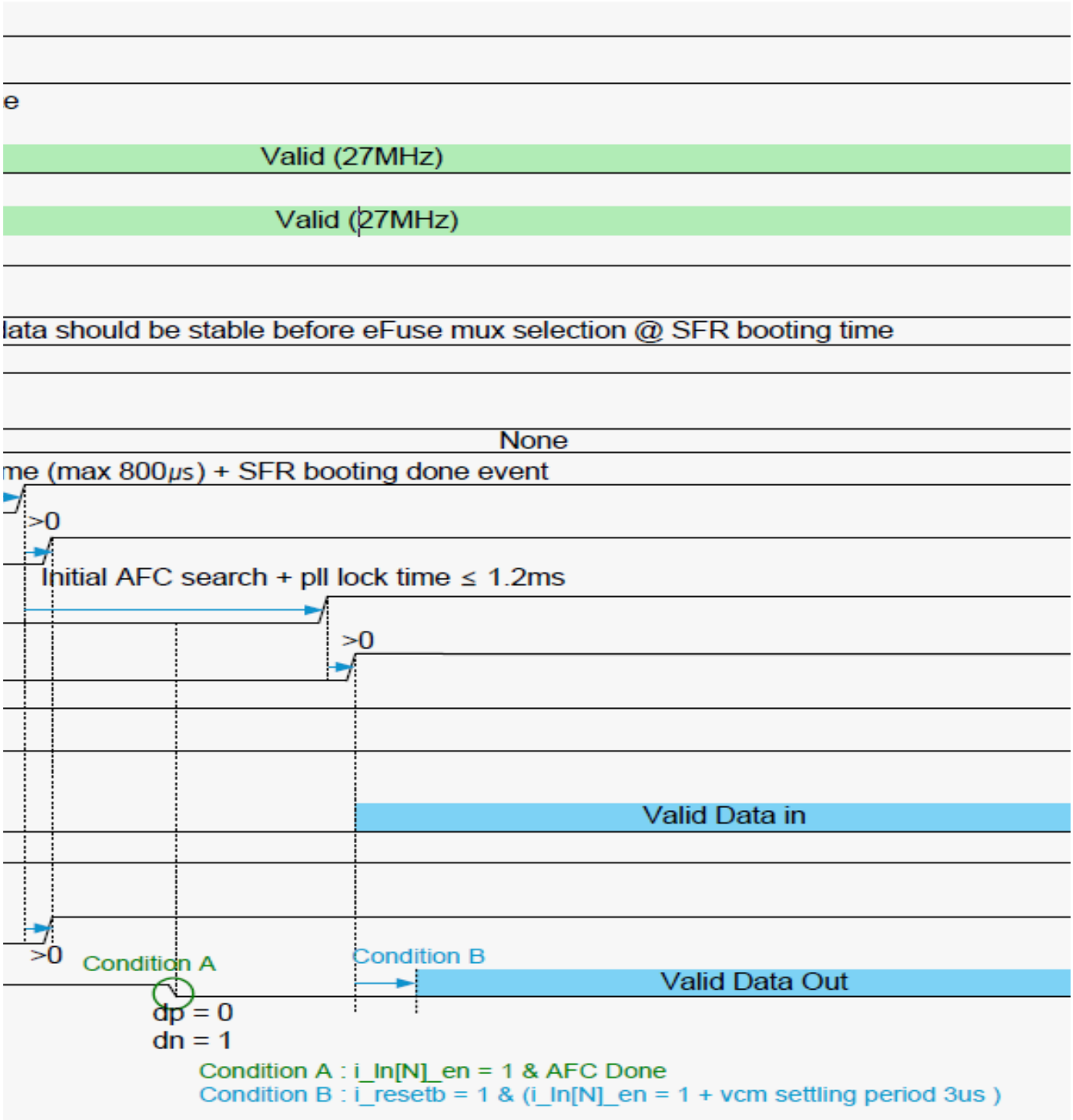
	설계 Spec 대비 Verificaiton결과 기록
	설계 Spec 대비 Verificaiton결과 기록

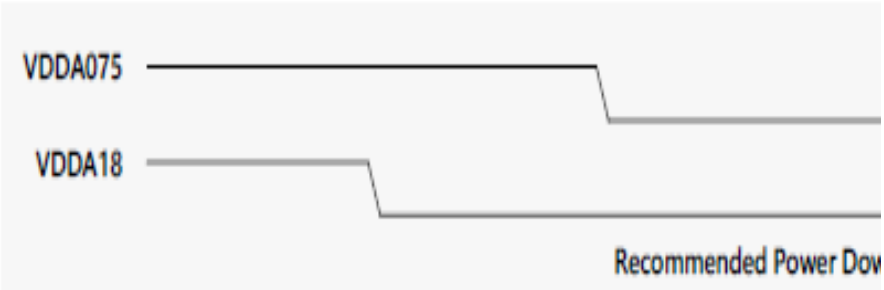
POWER UP SEQUENCE



- i_resetb_pll high, after at least 500us, o_tx_pll_lock high
- after i_resetb high, valid data is entered, after 2us, valid data is out.


e
Valid (27MHz)
Valid (27MHz)
lata should be stable before eFuse mux selection @ SFR booting time
None
me (max 800μs) + SFR booting done event
>0
Initial AFC search + pll lock time ≤ 1.2ms
>0
Valid Data in
>0
Condition A
Condition B
Valid Data Out
dp = 0
dn = 1
Condition A : i_In[N]_en = 1 & AFC Done
Condition B : i_resetb = 1 & (i_In[N]_en = 1 + vcm settling period 3us)

POWER DOWN SEQUENCE



vn sequence.

Low Power Moe in VBP

Normal mode + lane on/off : wake-up time 3us for each lane

Low power mode Type1 : wake-up time 3us

Low power mode Type2 : wake-up time 50us

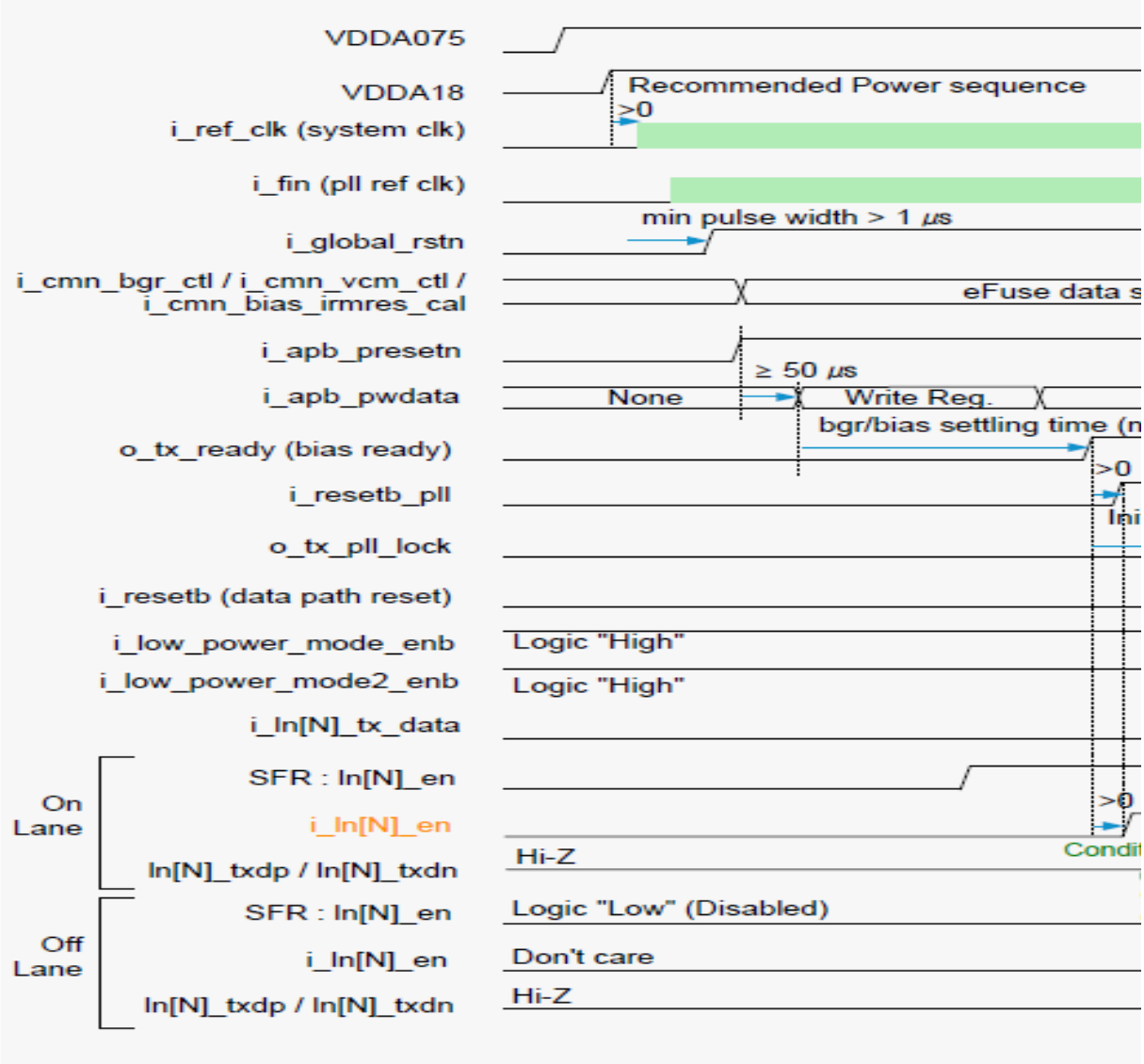
Low power mode Type3 : fast wake-up mode 200us, deep sleep mode 500us

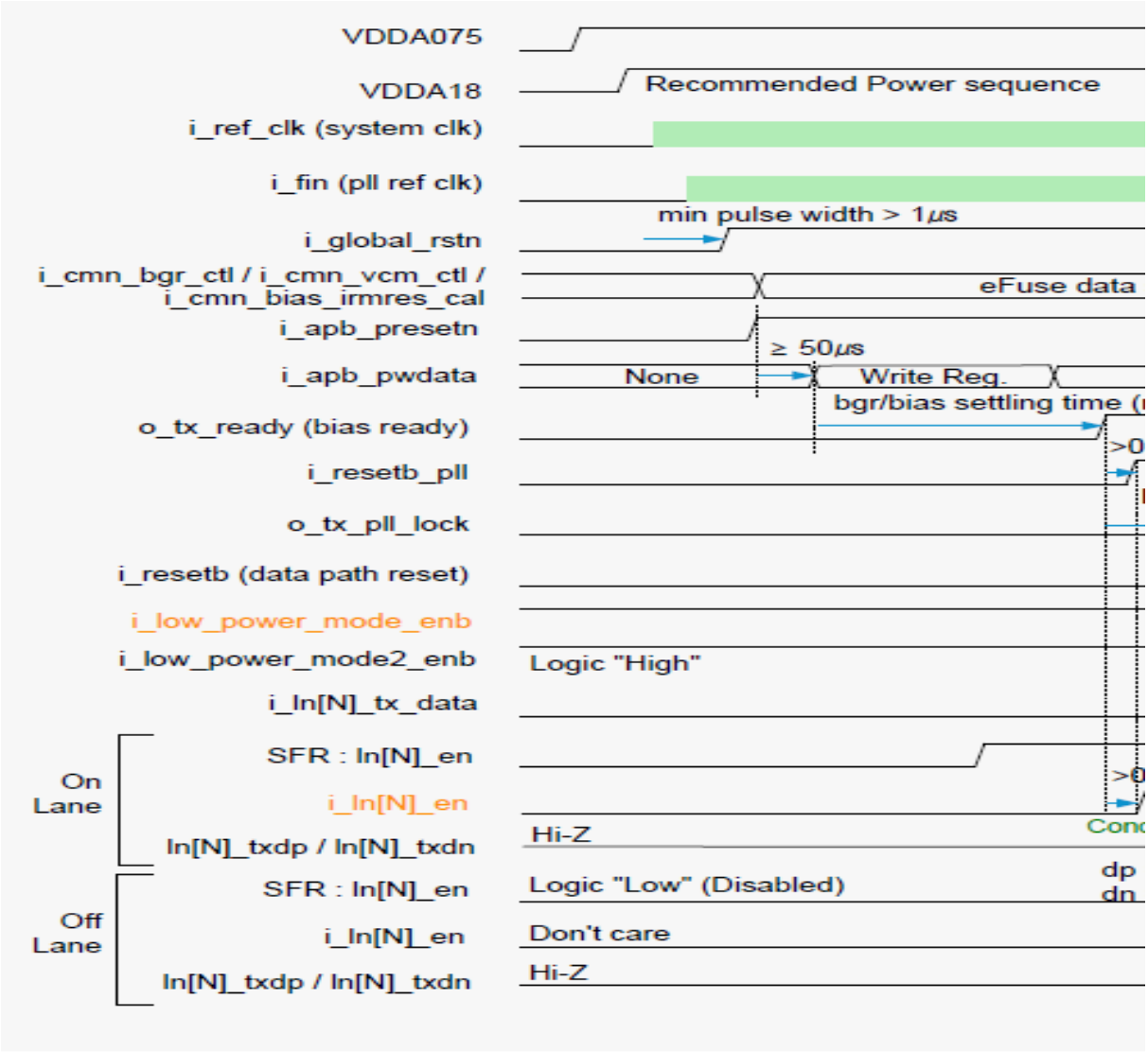
Power Mode		
	i_resetb_pll	i_
Normal mode + lane# on/off	1	
Low power mode Type1	1	
Low power mode Type2	0	
Low power mode Type3	1	

Power Mode			
	BIAS	PLL	PLL Ga
Normal mode + lane# on/off	ON	ON	
Low power mode Type1	ON	ON	
Low power mode Type2	ON	OFF	
Low power mode Type3	OFF	OFF	

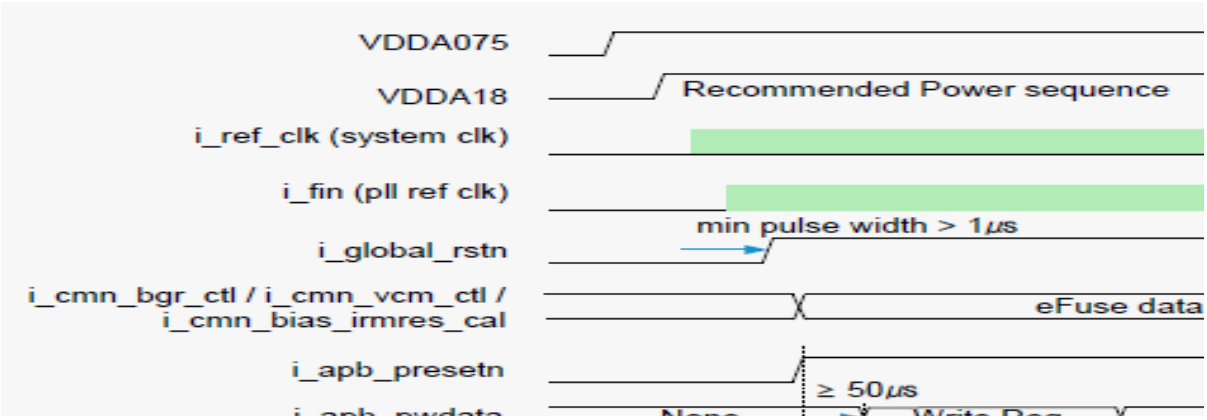
Low power mode Type1	Off	Off

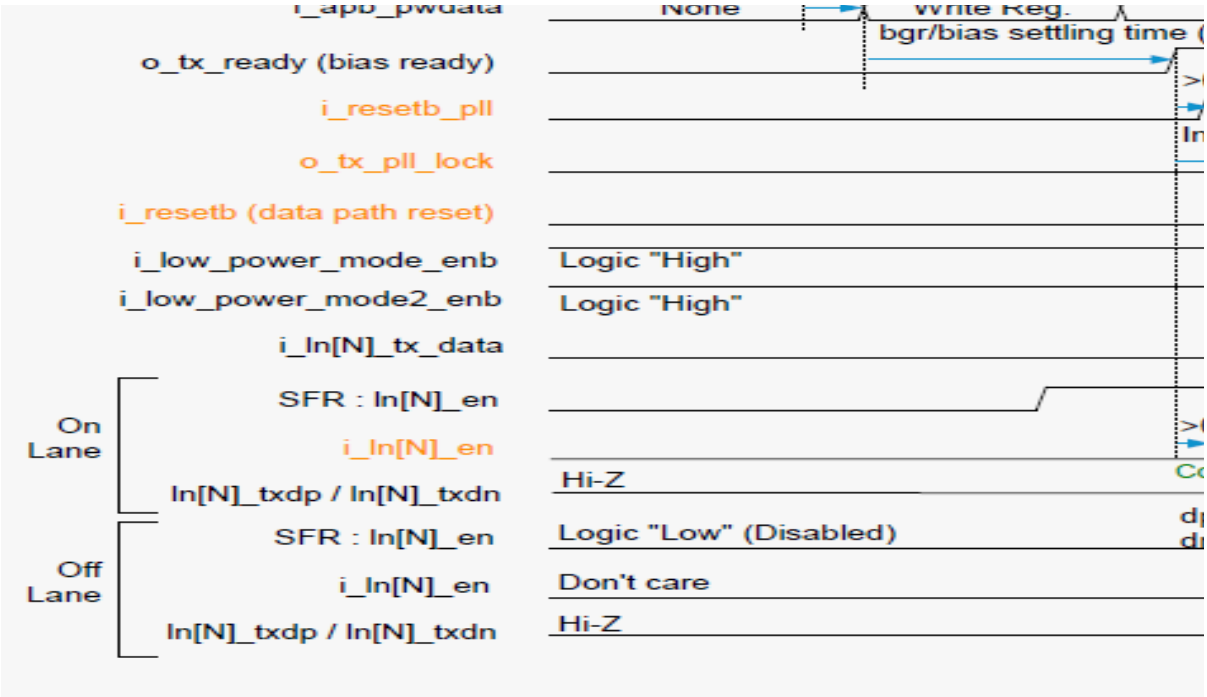
Normal mode + lane on/off



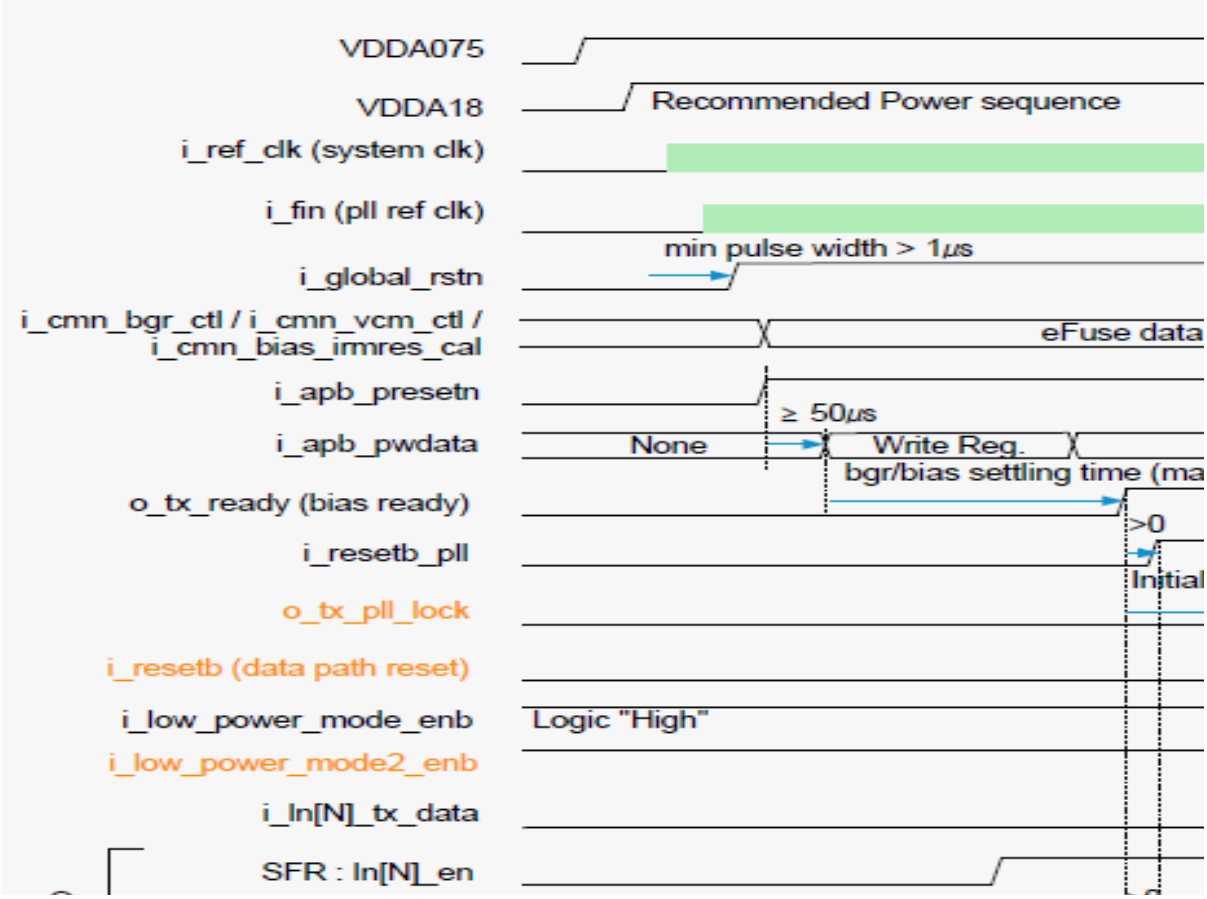


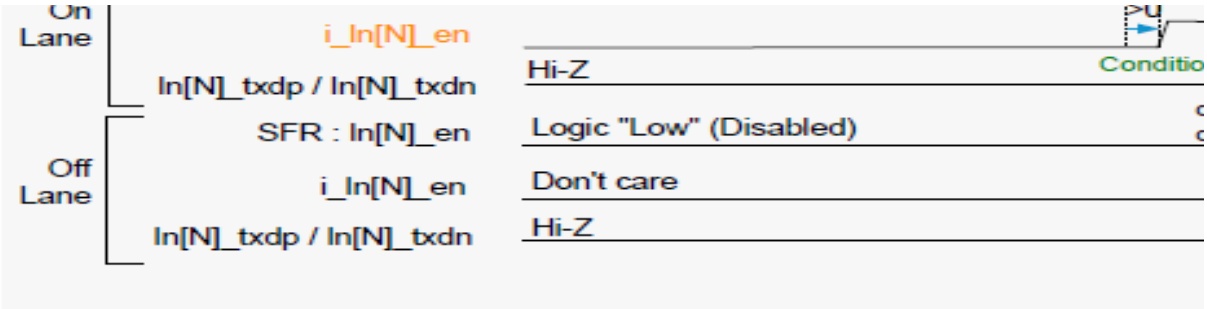
Low power mode Type2





Low power mode Type3

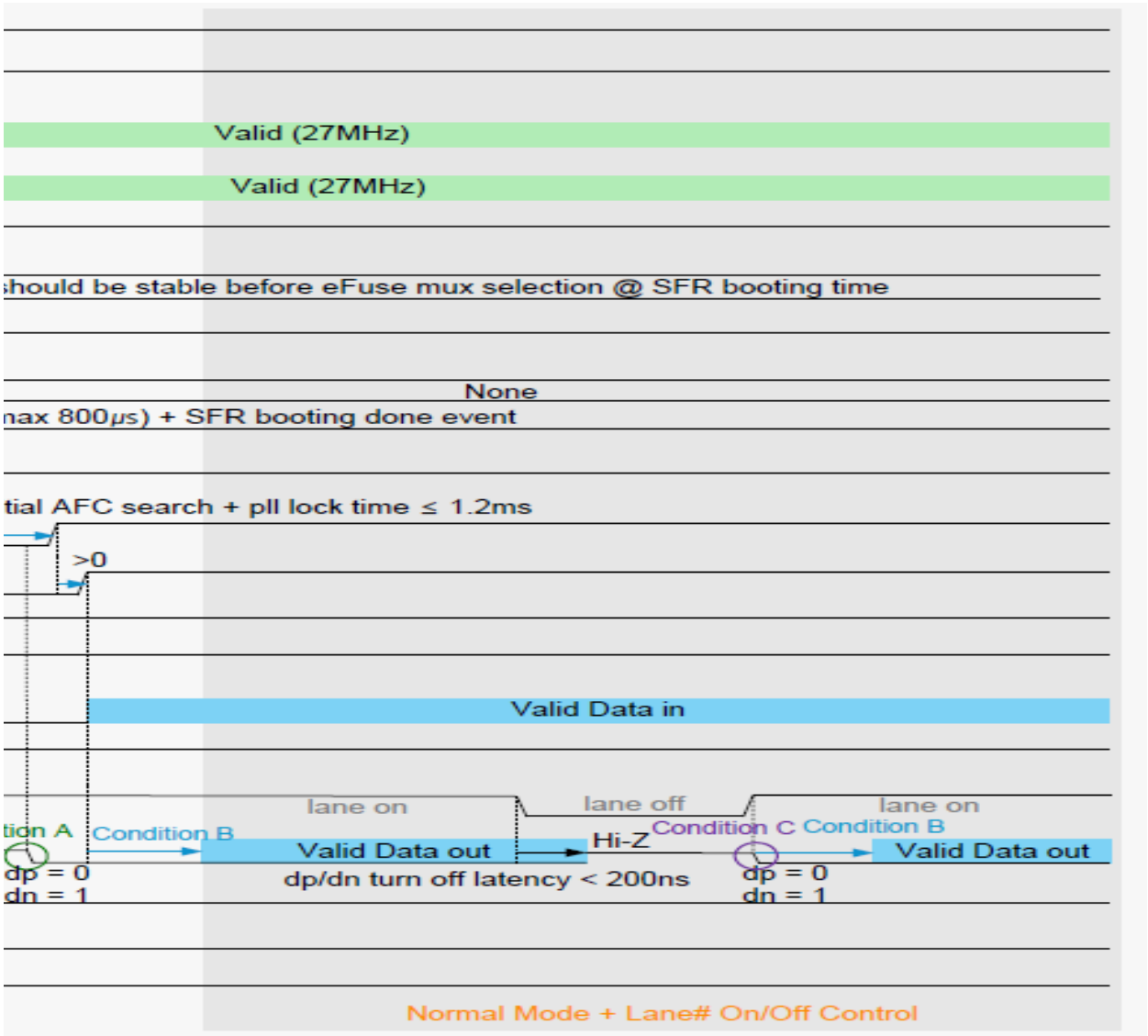



On Lane
i_In[N]_en
In[N]_txdp / In[N]_txdn
Hi-Z
Off Lane
SFR : In[N]_en
Logic "Low" (Disabled)
i_In[N]_en
Don't care
In[N]_txdp / In[N]_txdn
Hi-Z

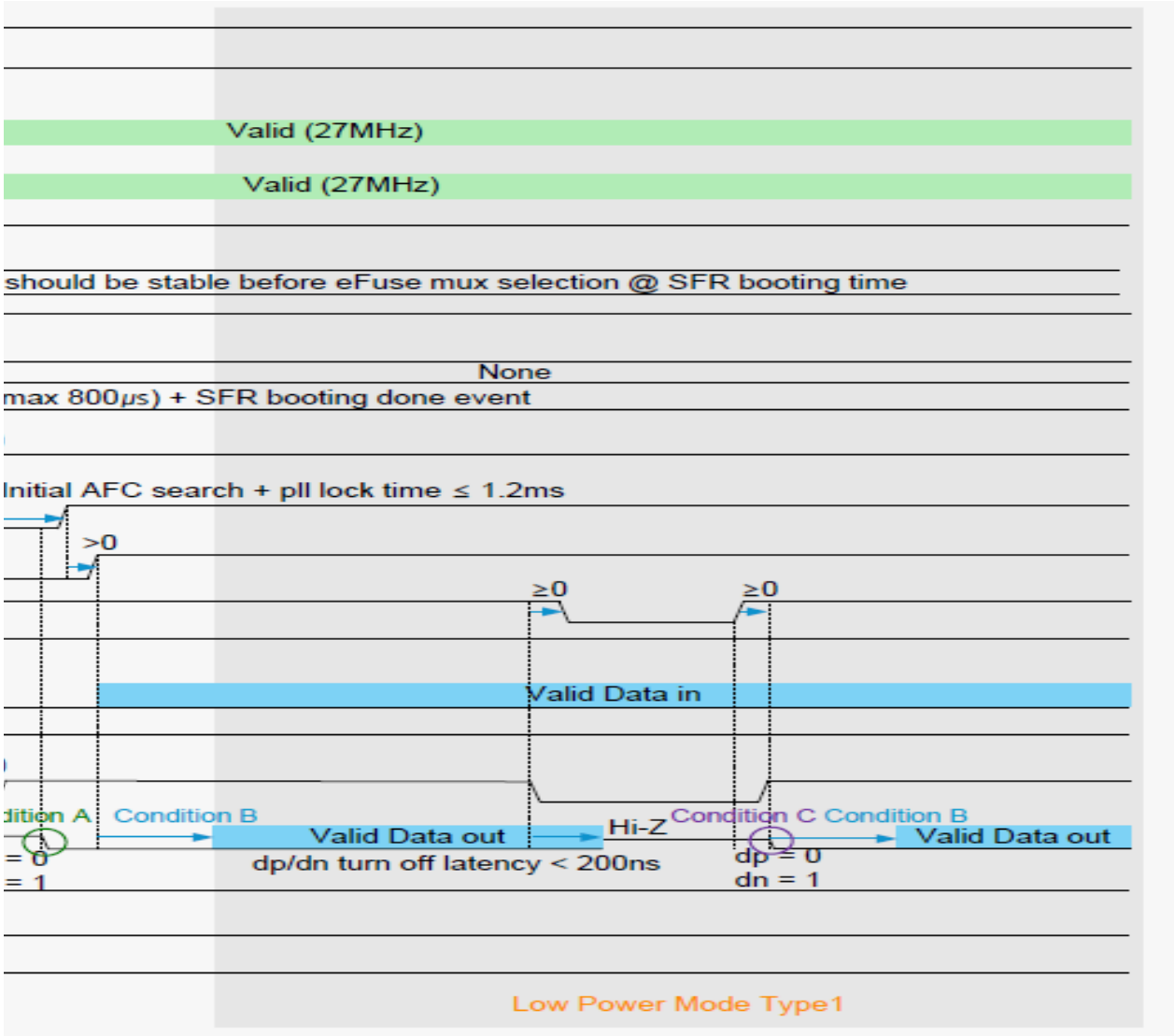
Input Ports			
_resetb	i_low_power _mode_enb	i_low_power _mode2_enb	i_In<N>_en (N : 0 ~ 15)
1	1	1	1 for enabled lane<N> 0 for disabled lane<N>
1	0	1	0 for all lanes
0	1	1	0 for all lanes
0	1	0	0 for all lanes

IP Status	
Clock ting	Tx Driver
X	• Enabled lanes: VCM on, P/N port: data • Disabled lanes: by o_SFR_low_power_tx_drv_mode[3:0] setting value – 4'b0001: VCM off, P/N port hi-Z (default) – 4'b0010: VCM off, P/N port hi-Z – 4'b0011: VCM on, P/N port DC-stuck at swing level – 4'b0100: VCM off, P/N port Tie on GND and VDD – 4'b0101: VCM off, P/N port hi-Z, Power Down Mode – 4'b0110~4'b1111: VCM on, P/N port DC-stuck at swing level **NOTE:** Lane on/off is configurable per lane.
O O O	• All disabled lanes: by o_SFR_low_power_tx_drv_mode[3:0] setting value – 4'b0001: VCM off, P/N port hi-Z (default) – 4'b0010: VCM off, P/N port hi-Z – 4'b0011: VCM on, P/N port DC-stuck at swing level – 4'b0100: VCM off, P/N port Tie on GND and VDD – 4'b0101: VCM off, P/N port hi-Z, Power Down Mode – 4'b0110~4'b1111: VCM on, P/N port DC-stuck at

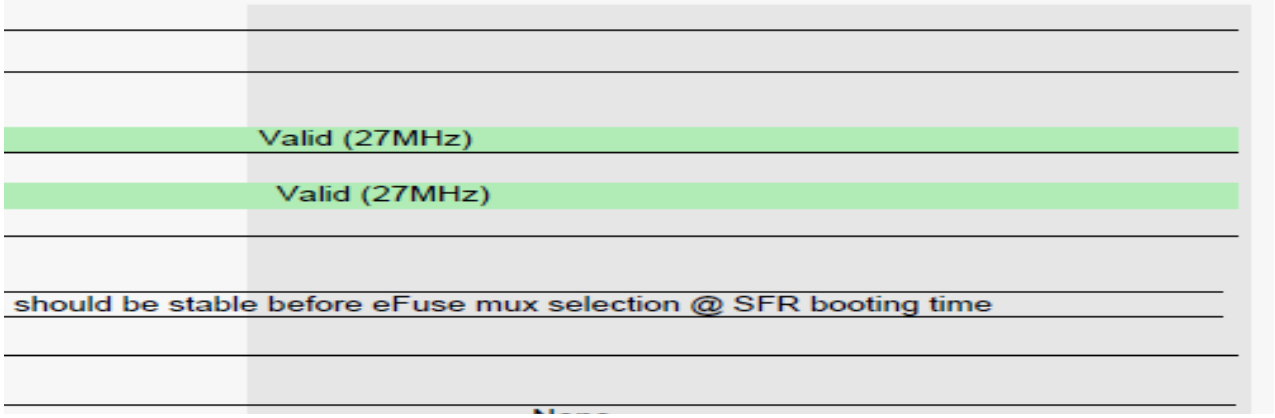
	swing level
NOTE:	All lanes are disabled in low power mode Type 1/2/3.

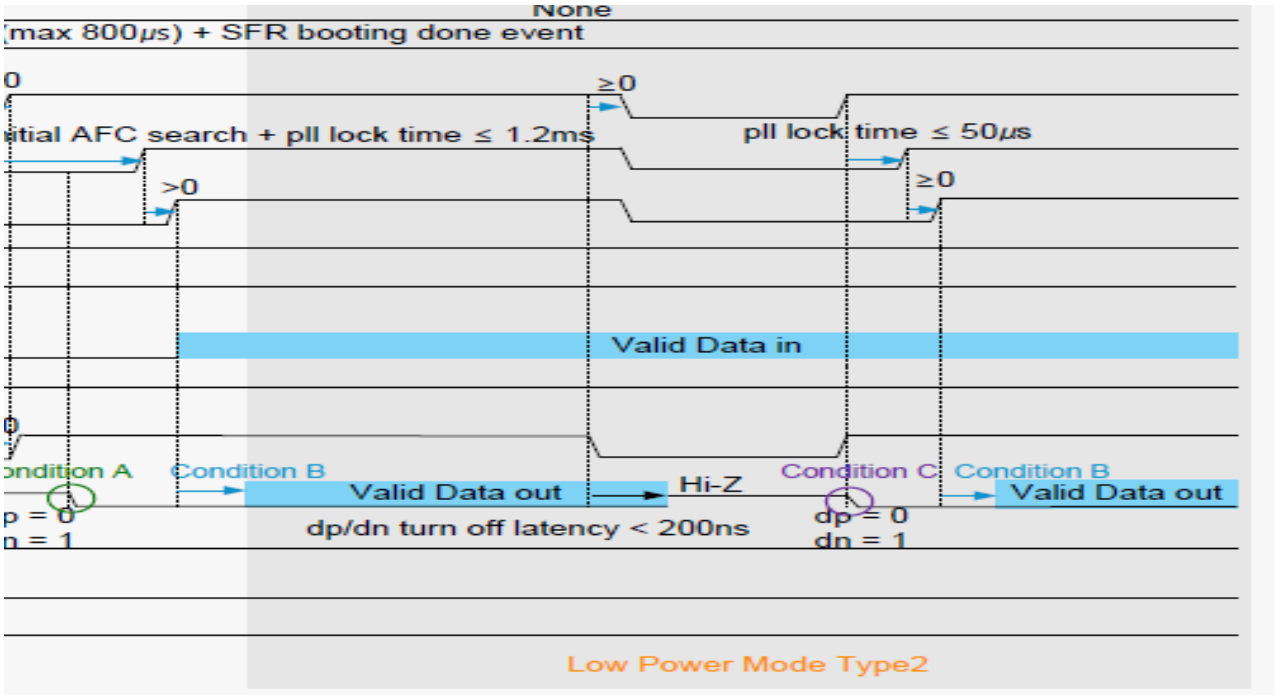


Condition A : i_ln[N]_en = 1 & AFC Done
Condition B : i_resetb = 1 & (i_ln[N]_en = 1 + vcm settling period 3us)
Condition C : i_ln[N]_en = 1

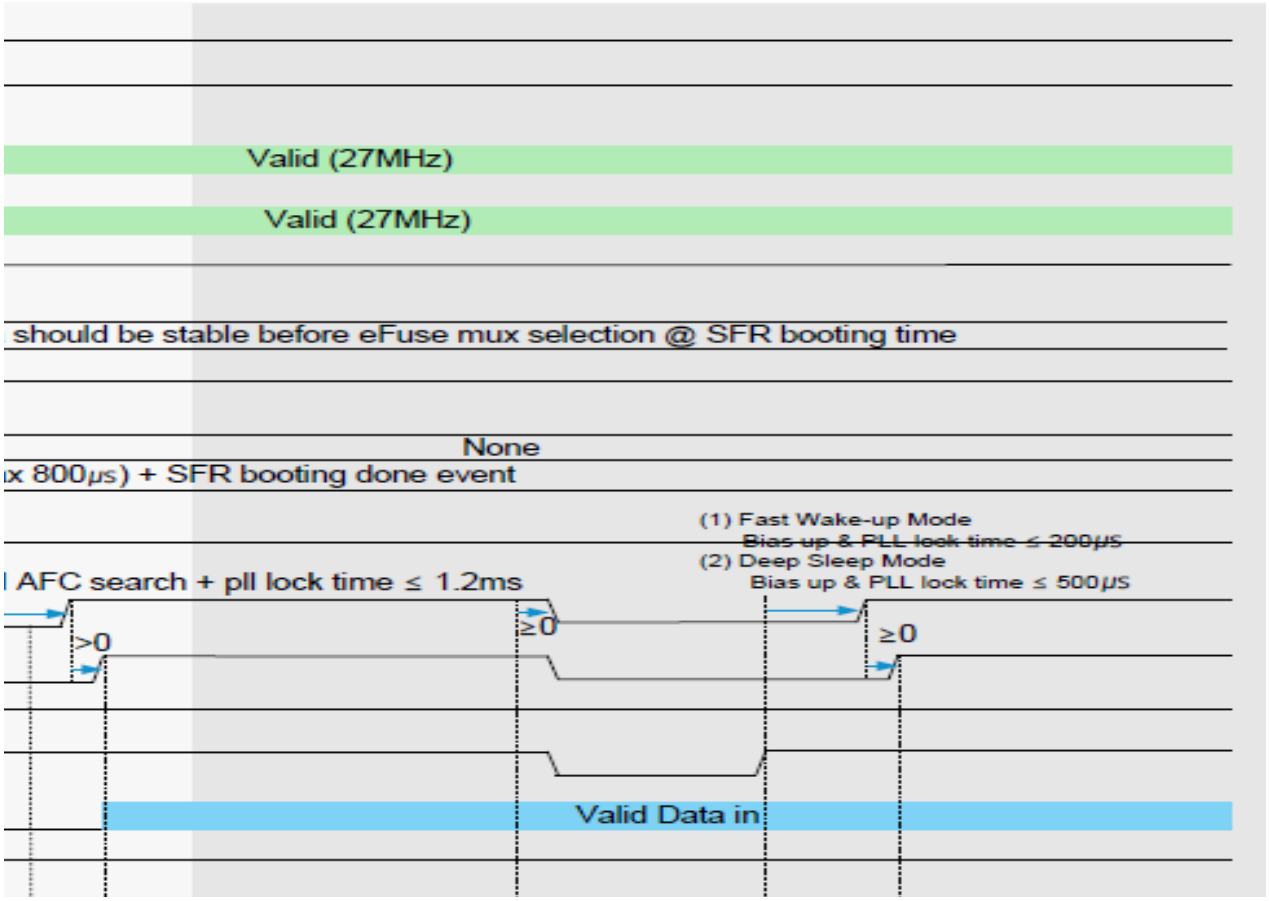


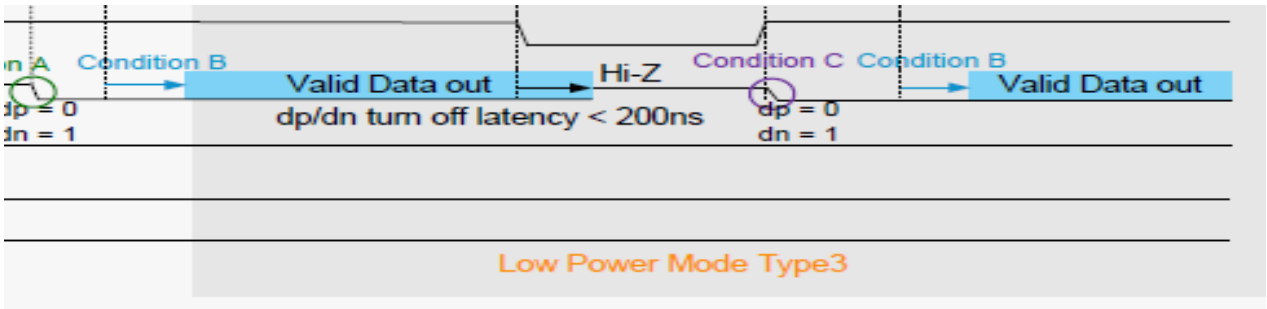
Condition A : i_In[N]_en = 1 & AFC Done
Condition B : i_resetb = 1 & (i_In[N]_en = 1 + vcm settling period 3us)
Condition C : i_In[N]_en = 1





Condition A : i_In[N]_en = 1 & AFC Done
Condition B : i_resetb = 1 & (i_In[N]_en = 1 + vcm settling period 3us)
Condition C : i_In[N]_en = 1





Condition A : i_In[N]_en = 1 & AFC Done
Condition B : i_resetb = 1 & (i_In[N]_en = 1 + vcm settling period 3us)
Condition C : i_In[N]_en = 1